JIYA SINGHAL

CS undergraduate, Class of 2027 | AI engineering: LLM products, evals and voice AI
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Experience

Eight Network **Oct 2026 – Present**
AI Software Engineer Intern *Bangalore, India*

- **AI Companion Chat App (primary engineer):** Built the **Fastify/Postgres** backend, **Flutter** app and React CMS for Simp, an app of 40 Hinglish personas with storylines and per-persona memory; Postgres alone prevents duplicate or missed messages (idempotent sends, per-conversation locks, SSE replay); runs on **Azure Container Apps** with GitHub Actions CI.
- **Grounded Astrology & Tarot:** Replaced invented readings with an in-house Vedic chart engine on NASA **JPL DE440s** ephemerides, validated on 64 frozen cases to within 0.5 arcsec, and seeded tarot draws with a card check, cutting invented cards from **16 of 33** turns to at most 1.
- **Model Gateway, Routing & Cost:** Routes each message type to Claude Sonnet, Opus or Haiku with an ordered fallback chain, per-call cost logging and per-user budget breakers; output guards drop cut-off or reasoning-leaked replies and retry once; about **$0.008 per turn**.
- **Evaluation Lab & LLM-as-Judge:** Regressions replay mocked on every merge; live runs use simulated users and pre-frozen criteria, judged blind pairwise by Opus in both orders. Found per-reply scores rising while conversation quality stayed flat; cut astrology mind-reading from **53% to 8%**.
- **Safety Guardrails:** Enforced no-meetup, no-contact and crisis-care rules plus an LLM check; evals showed most blocks were false alarms, so rebuilt it to count agreement only after a real ask to meet: **zero false positives and zero misses** on the labelled set; birth details only with consent.
- **Persona Convergence:** Traced personas sounding alike to engine rules outweighing each character prompt; cut the rule planner from **45 to 18** lines, kept safety in code, and decided on blind reads of 78-chat comparisons and 29k replayed replies.

Sing One Song **Aug 2025 – Oct 2026**
Software Engineering Intern *Bangalore, India*

- **Production Voice Pipeline:** Built the voice-onboarding backend (**FastAPI**, Firebase, idempotent Cloud Tasks) on GCP Cloud Run; ran its I/O-bound steps concurrently via `asyncio.gather()`, cutting measured cold end-to-end latency **57 s → 37 s**.
- **Recording Quality Gate:** Designed a rule-based gate on SNR, background dB, spectral flatness and temporal variance that blocks unusable recordings, cutting the warning rate **5.5% → 0.6%** and false warnings **∼11 → 0** on 355 real recordings.
- **Pitch-Model Migration:** Built a 21,750-case **Playwright** benchmark (29 singers × 150 noise files × 5 levels) that justified swapping 16MB PESTO for 389KB swift-f0: **40× smaller**, 98.8% vs 96.8% pass; kept as a regression check.
- **Real-Time DSP in Production:** Shipped true song transposition on web via a custom granular-SOLA **AudioWorklet**, keeping scoring in the key the singer hears, never falling silent; justified the CORS change with a written threat model.

Tradeindia **Jan 2025 – Apr 2025**
Product & Tech Intern *Noida, India*

- **Hybrid Product Search & Fraud Checks:** Combined **FAISS** semantic search with fuzzy string matching under a weighted scoring layer over unstructured seller listings, lifting search relevance **89% → 96%**; integrated Truecaller, IDfy and Shield identity verification into the buyer-seller flow.

Projects

Clinic Scheduling Agent with a Self-Improving Eval Loop | ***Python, Gemini function calling, Claude Opus judge***

- Designed a scheduling agent whose typed state changes only through tool results, with identity, slot ownership and idempotent booking enforced in tools, not the prompt; a loop adds one prompt rule per round behind a regression gate: **0.930 to 0.985** mean on 12 scripted scenarios, and a held-out simulated patient now passes.
- Audited the judge for position, length and self-preference bias: swapping answer order changed **3 of 12** verdicts, padded copies never won, and it marked down the read-back step the loop added on purpose, a case for gating on trace checks.

AI Persona: Voice + Chat Agent | ***FastAPI, Gemini, Vapi, ChromaDB, Voyage AI, Cal.com, Next.js***

- Shipped a grounded **RAG** agent over resume, notes and GitHub repos that answers by voice (∼1 s to first token) or streaming chat and books meetings via Cal.com tool calls; an LLM judge found **0 of 30** answers hallucinated, adversarial questions included.
- Red-teamed the RAG path with 10 poisoned documents; one leaked a test secret via a markdown link. Marking retrieved text as data, scanning chunks and filtering replies cut working attacks to **0 of 10** with 0 false blocks on 30 normal questions.

voicequal: Open-Source Speech & Singing Quality Library | ***Python, numpy/scipy, ONNX, MCP, PyPI***

- Published an audio-quality library on **PyPI**: replaced its first spectral SNR, off by 28 dB on noisy sung vowels, with an HNR + energy-SNR estimator (**4.2 dB** error) and distilled Microsoft's DNSMOS noise score into a 13k-parameter MLP; exposed as an MCP tool.

Education

Scaler School of Technology **July 2023 – 2027 (expected)**
Undergraduate Program in Computer Science CGR: ***7.22*** *Bangalore, India*

Birla Institute of Technology and Science (BITS), Pilani **July 2023 – 2027 (expected)**
B.Sc. (Hons.) in Computer Science CGPA: ***7.89***

Technical Skills

Languages	Python, TypeScript / JavaScript, Dart, SQL, Java
LLM Engineering	Claude / Gemini APIs, function calling, RAG (ChromaDB), model routing, prompt caching, MCP
Evals & Safety	LLM-as-judge evals, A/B testing, regression gates, guardrails, prompt-injection red teaming
Backend & Cloud	Node.js (Fastify), FastAPI, Postgres, Azure Container Apps, AWS, GCP Cloud Run, Docker
Mobile & Web	Flutter (Bloc), React, Next.js, Firebase (FCM, Firestore), SSE
Audio, Search & Testing	AudioWorklet, DSP (SOLA), pitch models (swift-f0), ONNX, FAISS, Playwright, pytest

Achievements

- **LeetCode Knight** (top ∼5%); **Open Source:** merged PRs to NixOS and OneBusAway; **Senior Diploma in Bharatanatyam**.